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Deliverable 1: Source Research Report

AI, Technology, Research & Venture Intelligence Dashboard — Data Layer Feasibility Study

Prepared as: Technical Source Research (Consulting-Grade) **Primary Model**
Dependency: Google Gemini API **Cost Posture:** Free-first; paid only where no viable free/RSS/API alternative exists **Date of Research:** July 2026 **Status:** Deliverable 1 of 3 (precedes Information Architecture and Technical Architecture)

A Note on Methodology and Epistemic Honesty

Before the category research, three ground rules that shaped every entry in this report:

1. **Verification over recall.** Pricing, rate limits, and API availability for data-source markets (funding intelligence, social platforms, news aggregation) change every few months — sometimes overnight. Every fact in this report that is time-sensitive (a price, a rate limit, a “does this API still exist”) was checked against live sources in July 2026, not pulled from static training knowledge. Facts that are structurally stable (e.g., “arXiv publishes an OAI-PMH feed,” “GitHub has a REST API”) are stated with more confidence because they have been stable for over a decade.
2. **Explicit uncertainty.** Where current pricing pages are ambiguous, contradictory, or gated behind “contact sales,” this report says so directly rather than inventing a number. You will see “Not publicly disclosed” in several places — that is itself a data point (it signals an unfavorable-to-indie-developer commercial posture).
3. **No invented APIs.** If a well-known product does **not** have a public API (e.g., LinkedIn, most VC firm websites), this report says so plainly and routes you to RSS/scraping-limits/manual-curation alternatives instead of fabricating an endpoint.

Source Evaluation Framework

Every source in this report is scored against the same rubric so that Deliverable 2 can programmatically rank and select sources:

Dimension	Scale
Reliability	High / Medium / Low (uptime, data accuracy, editorial trust)
Technical Difficulty	Low / Medium / High (auth complexity, format parsing, rate-limit handling)
Cost	Free / Freemium / Paid-only
Update Frequency	Real-time / Hourly / Daily / Weekly / Irregular
Long-term Scalability	Stable / At-risk / Actively degrading (e.g., platforms tightening API access)

Source-preference hierarchy applied throughout (as mandated): Official API → RSS → Official Website → Structured Public Dataset → Public GitHub Repo → Legal Scraping (only where ToS explicitly allow).

CATEGORY 1 — AI NEWS

1.1 Tier-1 Sources (Must-Have)

OpenAI Blog

- **Website:** <https://openai.com/news/> | **RSS:** Not officially published; third-party RSS bridges (e.g., RSS.app, FetchRSS) can wrap the page, but this is unofficial and fragile.
- **Official API:** No dedicated news API. **API Key Required:** N/A. **Login Required:** No.
- **Free Tier:** Fully free to read. **Paid Tier:** None for reading.
- **Update Frequency:** Irregular (announcement-driven, several times/month).
- **Reliability:** High (primary source). **Technical Difficulty:** Medium (requires scraping or an unofficial RSS bridge since there's no native feed).
- **Integration Method:** HTML scraping (allowed for personal/aggregation use in most jurisdictions when respecting robots.txt) or a third-party RSS proxy.
- **Data Quality:** Very High. **Geographic Coverage:** Global. **Historical Data:** Yes, full archive on-site.
- **Recommended:** Yes. **Why:** Primary-source model/product announcements; zero cost.
- **Risks:** No RSS = fragile scraping; site structure changes break parsers periodically.
- **Long-term Scalability:** Stable as a source of truth; scraping layer needs maintenance.

Anthropic News

- **Website:** <https://www.anthropic.com/news> | **RSS:** Not officially published as of this research.
- **API:** None for blog content. **Login Required:** No.
- **Update Frequency:** Irregular, several posts/month.
- **Reliability:** High. **Technical Difficulty:** Medium (scrape or third-party RSS bridge).
- **Recommended:** Yes — primary source for Claude model announcements, safety research, and policy positions.
- **Risks:** Same scraping fragility as OpenAI.

Google DeepMind Blog

- **Website:** <https://deepmind.google/discover/blog/> | **RSS:** Google blogs historically expose Atom/RSS feeds via Blogger infrastructure on some properties; DeepMind's current site is a custom CMS without a confirmed public feed — **verify at implementation time.**
- **Recommended:** Yes, primary source for Gemini/DeepMind research releases.

Meta AI Blog

- **Website:** <https://ai.meta.com/blog/> — no confirmed public RSS as of this research.
- **Recommended:** Yes for Llama/model releases; scrape or monitor via GitHub release pages (more reliable, see Category 2).

TechCrunch (AI Section)

- **Website:** <https://techcrunch.com/category/artificial-intelligence/>
- **RSS: Yes, official** — <https://techcrunch.com/category/artificial-intelligence/feed/> (TechCrunch publishes category-level RSS feeds; this is one of the most reliable free structured feeds in the entire report).
- **API Key Required:** No. **Login Required:** No. **Free Tier:** Full feed, free, unlimited for personal/aggregation use.
- **Update Frequency:** Multiple times/hour during business hours.
- **Reliability:** High. **Technical Difficulty:** Low (standard RSS/Atom parsing).
- **Data Quality:** High, professional editorial. **Historical Data:** Feed shows recent items only; full archive via website.
- **Recommended: Yes** — **Best Free Source for this category.**
- **Risks:** Feed occasionally throttled/changed without notice; keep a fallback scraper.

The Verge / Ars Technica / Wired (AI sections)

- All three publish standard category-level RSS feeds (e.g., theverge.com/rss/index.xml, arstechnica.com/feed/). Free, no key, no login. High reliability, high technical ease. Recommended as breadth-of-coverage supplements to TechCrunch.

Hacker News (as an AI-news aggregator)

- See full profile in Category 5 (Social Intelligence) — also functions as a de facto crowd-curated AI news filter via its Algolia-backed **official, free, keyless API**: <https://hn.algolia.com/api>.

AI-Focused Newsletters (Substack-hosted)

- **Examples:** Import AI (Jack Clark), The Batch (DeepLearning.AI), Ben's Bites, TLDR AI.
- **RSS:** Substack **natively exposes RSS for every publication** at `{publication}.substack.com/feed` — this is a durable, official, free mechanism independent of any individual newsletter's goodwill.
- **API:** No general Substack public API for content; RSS is the correct integration path.
- **Free Tier:** Yes, RSS is free regardless of the newsletter's paid-subscription tiers (paid tiers usually gate a subset of posts, not the RSS existence itself — verify per-publication, some restrict RSS content to teaser-only for paid posts).
- **Recommended: Yes** — **Best RSS mechanism in the entire AI News category.**
- **Risks:** Editorial curation bias per author; content overlap across newsletters.

1.2 Category 1 Summary Table

Source	API	RSS	Login	Cost	Reliability	Difficulty	Recommended
OpenAI Blog	No	Unofficial only	No	Free	High	Medium	Yes
Anthropic	No	Unofficial	No	Free	High	Medium	Yes

Source	API	RSS	Login	Cost	Reliability	Difficulty	Recommended
News		only				m	
Google DeepMind Blog	No	Unconfirmed	No	Free	High	Medium	Yes
Meta AI Blog	No	Unconfirmed	No	Free	High	Medium	Yes
TechCrunch AI	No	Yes (official)	No	Free	High	Low	Yes — Best Overall
The Verge / Ars Technica / Wired	No	Yes (official)	No	Free	High	Low	Yes
Hacker News (Algolia API)	Yes (official, free)	Yes	No	Free	High	Low	Yes
Substack AI newsletters	No	Yes (official, per-pub)	No	Free	Medium-High	Low	Yes
MIT Technology Review AI	No	Yes (official)	Paywall for full text	Freemium	High	Low	Good to Have

Category 1 Recommendations - Best Source: TechCrunch AI RSS (breadth + reliability + zero friction). - **Best Free Source:** Same — TechCrunch RSS. - **Most Reliable:** Official company blogs (primary source, but weak machine-readability). - **Easiest Integration:** RSS feeds generally (TechCrunch, Verge, Substack). - **Best API:** Hacker News Algolia API (structured, free, no key). - **Best for MVP:** RSS aggregation of TechCrunch + Verge + Ars Technica + 3-4 curated Substack newsletters + HN Algolia API, all fed through Gemini for summarization/dedup. - **Best for Future Scaling:** Add a proper news-aggregation API (see paid options below) once budget allows, to reduce scraper maintenance burden. - **Overall Recommendation:** Build the MVP entirely on RSS + the HN API. Avoid paid news APIs initially (see cost findings below).

1.3 Paid News Aggregator Landscape (context, not recommended for MVP)

Research into general-purpose news APIs (NewsAPI.org, NewsAPI.ai/Event Registry, GNews, NewsData.io, Mediastack, Webz.io) surfaced a consistent pattern relevant to this project’s “prefer free” constraint: news API pricing ranges from free with significant limitations to \$500+/month for enterprise plans, with the most common production plans

costing \$49–249/month, and free tiers typically capped at 100–200 requests per day with commercial-use restrictions. Specifically, NewsAPI.org’s free tier is restricted to development/localhost use, requiring a \$449/month Business plan for any production deployment. Given that this dashboard needs production, continuous ingestion, **no general news API is recommended for the MVP** — RSS coverage of the specific outlets that matter (TechCrunch, Verge, Ars Technica, newsletters) is both free and sufficient, and avoids the “pay for 150,000 irrelevant sources to get 20 relevant ones” problem these aggregators represent.

CATEGORY 2 — AI MODEL RELEASES

The single most reliable, lowest-maintenance channel for model-release tracking across nearly every major lab is **GitHub Releases + Hugging Face Model Hub**, not company blogs. This is a key architectural finding for Deliverable 2.

Organization	Best Official Channel	API/RSS	Notes
OpenAI	Blog + platform.openai.com/docs/models (model index page)	No RSS; no models API listing changelog	Track via blog scraping + Hugging Face community trackers
Anthropic	https://www.anthropic.com/news + docs.claude.com model pages	No RSS confirmed	Model ID strings visible via the Anthropic API’s /v1/models list endpoint (official, free, requires API key)
Google DeepMind / Gemini	https://deepmind.google/models/ + Google AI blog	No confirmed RSS	Also announced via Google Cloud release notes RSS

Organization	Best Official Channel	API/RSS	Notes
			(cloud.google.com publishes release-note feeds for many products)
Meta	https://ai.meta.com/blog/ + GitHub org github.com/meta-llama	GitHub org has an official, free, structured releases/commits API	Recommended primary channel — GitHub Releases API is the most machine-readable option in this entire category
Microsoft	https://blogs.microsoft.com/ai/ + Azure AI updates page	Azure has official RSS for “What’s new” pages	Good to Have
AWS	https://aws.amazon.com/blogs/machine-learning/	AWS “What’s New” has an official RSS feed: aws.amazon.com/about-aws/whats-new/recent/feed/	Recommended — official, structured, free
NVIDIA	https://blogs.nvidia.com/ + NVIDIA Developer blog	Developer blog has RSS	Good to Have
Mistral AI	https://mistral.ai/news	No confirmed RSS; also on GitHub (github.com/mistralai)	GitHub releases recommended
xAI	https://x.ai/news	No RSS confirmed	Low-frequency source; monitor manually or via HN/social

Organization	Best Official Channel	API/RSS	Notes
Cohere	https://cohere.com/blog	No RSS confirmed	mentions Lower priority
Perplexity	https://www.perplexity.ai/hub/blog	No RSS confirmed	Lower priority for “model releases” (Perplexity is mostly a product, not a foundation-model lab)
Databricks	https://www.databricks.com/blog	Has RSS via blog category feeds	Relevant for MosaicML/DBRX-lineage releases
Snowflake	https://www.snowflake.com/en/blog/	Has RSS	Relevant for Snowflake Arctic-lineage releases
Hugging Face	https://huggingface.co/blog + Hub API	Official, free, keyless Hub API: https://huggingface.co/api/models?sort=createdAt&direction=-1 returns newly published models across <i>every</i> organization publishing to HF, sortable by date, downloads, likes	This is the single best structured source in the whole category — it captures releases from OpenAI-compatible community ports, Meta, Mistral, Alibaba/Q

Organization	Best Official Channel	API/RSS	Notes
			wen, DeepSeek, Google's open Gemma models, and hundreds of smaller labs, all in one machine-readable JSON endpoint.

Category 2 Recommendations

- **Best Source:** Hugging Face Hub API (/api/models) — free, keyless, structured, covers nearly every lab that ships open weights.
- **Best Free Source:** Same.
- **Most Reliable:** GitHub Releases API for organizations that ship code (Meta/Llama, Mistral) — official REST API, well-documented, generous free rate limits (see Category 9 for GitHub API specifics).
- **Easiest Integration:** Hugging Face Hub API — single GET request, JSON, no auth needed for public read.
- **Best API:** Hugging Face Hub API.
- **Best RSS:** AWS “What’s New” feed (only fully confirmed official RSS in this category).
- **Best for MVP:** Hugging Face Hub API (broad open-weight coverage) + GitHub Releases API for meta-llama, mistralai, google-deepmind, QwenLM, deepseek-ai orgs + manual/scraped tracking of OpenAI, Anthropic, and Google closed-model blogs (since these are lower-frequency, high-importance events where scraping fragility is an acceptable tradeoff).
- **Overall Recommendation:** Do not try to build 12 separate scrapers on day one. Start with the two structured APIs (HF Hub + GitHub) which will surface 80%+ of relevant model-release events, then layer in blog-scraping for the three or four closed-weight labs (OpenAI, Anthropic, Google, xAI) where it matters most.

CATEGORY 3 — STARTUP FUNDING INTELLIGENCE

This category has the least free-tier support in the entire report. Verified findings:

Crunchbase

- **Official API:** Yes, exists, but commercially gated. As of this research, Crunchbase’s own documentation states “Access to the Full API Requires an Enterprise or Applications License,” meaning the cheapest self-serve plans do not guarantee full API access at all — it is a licensing conversation, not a checkout flow.
- **Free Tier:** As of 2025, Crunchbase eliminated free API access entirely.
- **Paid Tier / Estimated Cost:** Self-serve UI plans run \$99/month for Pro (or \$49/month billed annually), and \$199/month for Business billed annually, but these are **UI/seat licenses, not API licenses**. Actual API access is quoted separately, and enterprise contract data suggests Enterprise contracts with API access typically start at \$50,000+ annually.
- **Recommended: Not Recommended for this project.** Cost is entirely incompatible with the ₹0 MVP constraint and remains expensive even at scale.
- **Free Alternatives:** See below.

Crunchbase Alternatives (Free/Low-Cost)

Source	API/RSS	Free Tier	Notes
Y Combinator (companies + Launch posts + blog)	No formal API; YC’s public company directory (ycombinator.com/companies) is scrapeable; YC’s Hacker News “Launch HN” and “Show HN” posts are captured for free by the HN Algolia API	Free	Excellent for early-stage/seed startup signal, YC-specific but very high signal-to-noise
TechCrunch funding-tagged RSS	Official RSS (techcrunch.com/tag/funding/feed / pattern)	Free	Strong for announced-round coverage, English-language/US-EU bias
AngelList / Wellfound	No confirmed public funding-data API in 2026 (Wellfound is primarily a jobs marketplace now); historical AngelList API for startups/investors has been deprecated in past years — treat as unavailable, verify at build time	N/A	Not recommended as a data source; useful only as a manually-browsed reference
Dealroom.co	Has enterprise API; public free access is limited to occasional published reports/datasets, not a general free API	Paid (enterprise)	Not recommended for MVP; revisit at scale
Tracxn	Enterprise SaaS product with API; no meaningful free tier identified	Paid	Not recommended for MVP
SEC EDGAR (Form D	Official, free, structured, keyless government API and full-text	Free	High-value, underrated source

Source	API/RSS	Free Tier	Notes
filings)	search (<code>efts.sec.gov/LATEST/search-index?q=...</code> and the EDGAR full-text search UI)		— every US private placement (most VC-backed rounds) files a Form D; this is a <i>primary legal document</i> , not press-release spin, and it is 100% free with no rate-limit gate for reasonable use
Government startup portals (see Category 8 for full detail)	Startup India, GEM (India), EU Startup portals	Free, official	Regional/government-verified startup registries
Crunchbase News (the editorial blog, distinct from the data API)	RSS available on the blog	Free	Good for narrative/analysis, not structured data
PitchBook	Enterprise-only, no meaningful free tier	Paid, high cost	Not recommended

Category 3 Recommendations

- **Best Source (data completeness):** Crunchbase — but commercially blocked for this project.
- **Best Free Source:** A combination of **SEC EDGAR Form D full-text search** (structured, legal-grade, free) + **TechCrunch funding RSS** (narrative coverage) + **HN Algolia API filtered for “Show HN”/“Launch HN”/funding keywords**.
- **Most Reliable:** SEC EDGAR — it is a US government system of record, not subject to a vendor’s commercial roadmap.
- **Easiest Integration:** TechCrunch RSS.
- **Best API:** SEC EDGAR full-text search API (free, official, structured, though US-only in scope).
- **Best for MVP:** SEC EDGAR (US coverage) + TechCrunch/VC-newsletter RSS (global narrative coverage) + Startup India portal (India-specific, see Category 8) + YC directory scraping. Explicitly **exclude Crunchbase, Dealroom, Tracxn, and PitchBook from MVP scope**.
- **Best for Future Scaling:** Crunchbase Pro/Business (\$49–199/month) becomes viable once the dashboard has proven value and budget exists; full API access still

requires an enterprise conversation even then, so evaluate Dealroom or Tracxn enterprise tiers against Crunchbase's enterprise tier at that point rather than assuming Crunchbase is the default upgrade path.

- **Risks:** US/EU-heavy free-source bias — India- and APAC-specific coverage will be comparatively thin unless government portals and regional newsletters are actively curated (see Category 8).

CATEGORY 4 — VENTURE CAPITAL INTELLIGENCE

Individual VC firms overwhelmingly do **not** expose APIs. The realistic, free integration surface is:

Source Type	Mechanism	Cost	Notes
VC firm blogs/partner blogs (a16z, Sequoia, Accel, Lightspeed, Bessemer, Elevation Capital, Blume Ventures, etc.)	RSS where available (many run on Substack, Ghost, or WordPress — all of which natively support RSS), else scraping	Free	Verify per-firm; a16z publishes on a16z.com with content also mirrored to Substack in some cases
Portfolio pages	No API; scraping only, low update frequency, low priority for automation	Free	Better suited to periodic manual/batch refresh than real-time polling
Fund announcement press releases	Captured indirectly via TechCrunch/VC-newsletter RSS and Form D filings (Category 3)	Free	Avoid building a dedicated scraper per fund; let news RSS + EDGAR do the work
Accelerators/Incubators (YC, Techstars, 500 Global, Antler, Sequoia Surge, IIM/IIT incubation cells)	Mixed: YC has a scrapeable directory; most others are website-only	Free (scraping)	Batch/periodic refresh recommended
Angel Networks (Indian Angel Network, LetsVenture, AngelList syndicates)	No public APIs identified	Free (manual monitoring)	Low priority for MVP automation
Government VC funds (Fund of Funds for Startups India - FFS, SIDBI, In-Space, EU's InvestEU)	Official government sites, some with RSS (see Category 8)	Free, official	High-trust, low-frequency source

Category 4 Recommendations

- **Best Source:** VC-focused newsletters (Substack-native RSS) — highest signal-to-effort ratio.
 - **Best Free Source:** Same.
 - **Most Reliable:** Government VC fund pages (FFS/SIDBI, InvestEU) for policy-grade accuracy, though low frequency.
 - **Easiest Integration:** Substack RSS from VC partner newsletters.
 - **Best API:** None exists natively for this category; **Best for MVP:** none required — this category is fundamentally an RSS + periodic-scrape category, not an API category.
 - **Overall Recommendation:** Do not over-invest engineering time here at MVP stage. Curate a list of ~15–20 high-signal VC/partner Substacks and blogs with RSS, aggregate via Gemini for thematic tagging (e.g., “which funds are actively writing about agentic AI”), and treat portfolio-page scraping as a v2 feature.
-

CATEGORY 5 — SOCIAL INTELLIGENCE

This is the most legally and commercially constrained category. Findings by platform:

X (Twitter)

- **Official API:** Exists but is **paid-only since 2023**; no meaningful free tier for read access at the volume a dashboard would need. **Not recommended.**
- **Free Alternative:** None reliable and compliant. **Recommendation: exclude X from the automated MVP pipeline**; rely on X mentions surfaced secondhand through TechCrunch/HN/newsletter coverage instead.

LinkedIn

- **Official API:** Exists only for narrow, partner-approved use cases (e.g., “Sign in with LinkedIn,” limited marketing APIs); there is **no general-purpose public content-read API** for company/post monitoring. Scraping LinkedIn violates its Terms of Service and has been the subject of prior litigation (hiQ v. LinkedIn and related cases show this is legally contested territory, not a safe default).
- **Recommendation: Not Recommended.** Exclude from the automated pipeline entirely; do not scrape.

GitHub

- **Official API:** Yes — REST and GraphQL, free, generous limits (5,000 requests/hour authenticated, 60/hour unauthenticated — standard, well-documented, stable for years).
- **Recommended: Yes — one of the best sources in the entire report.** Full profile in Category 9.

Reddit

- **Official API:** Yes, exists, but has become progressively more restrictive. As of this research: the free, non-commercial tier is capped at 100 queries per minute per OAuth client, while commercial use requires a paid, manually-approved agreement billed at roughly \$0.24 per 1,000 API calls. Separately, since late 2025 Reddit routes new OAuth token requests through a manual approval process, and developers report rejections even for small, clearly-described personal projects.
- **Free Tier:** Usable for non-commercial personal/research use within 100 QPM, **if approval is granted** — approval is not guaranteed.
- **Recommendation: Conditional.** Suitable for a personal-use, non-commercial dashboard (which this project is, per the brief) via PRAW + free OAuth tier, but budget engineering time for the approval-application process and build a graceful-degradation path (e.g., skip Reddit ingestion silently) in case approval is denied or later revoked.
- **Risk:** Reddit's trajectory (2023 API lockdown → 2025 manual-approval wall) indicates continued tightening; **do not architect the dashboard to depend critically on Reddit.**

Hacker News

- **Official API:** Yes — two options: (1) the **official Firebase API** (`hacker-news.firebaseio.com`), free, keyless, real-time; (2) the **Algolia-powered search API** (`hn.algolia.com/api`), free, keyless, full-text searchable, better for “search for AI-related stories” use cases.
- **Free Tier:** Fully free, no rate-limit gate for reasonable personal-project use.
- **Recommended: Yes — Best Source in this entire category.** No login, no key, no cost, high signal for developer/AI/startup discourse.
- **Risks:** None material identified.

Substack

- Covered in Category 1. RSS-native, free, official, durable.

YouTube

- **Official API:** Yes — YouTube Data API v3, free tier with daily quota (historically 10,000 “quota units”/day, enough for moderate channel-monitoring use, e.g., tracking new uploads from AI-lab channels). Requires a Google Cloud API key (free to create).
- **Recommended:** Yes, for tracking official channel uploads (OpenAI, Google DeepMind, Anthropic, major AI YouTube educators) — **verify current quota figures at implementation time**, as Google adjusts quotas periodically.

Discord / Slack

- **Official API:** Both exist but are designed for building bots *within* servers you have been invited to and granted permissions in — not for broad external monitoring of arbitrary public communities without an invite and bot-install step per server.

- **Recommendation:** Low priority for MVP; treat as a v2 feature requiring manual per-community bot installation, not a general “source.”

Category 5 Summary Table

Source	API	Cost	Login/Approval	Reliability	Recommended
X (Twitter)	Paid only	Paid	Yes	Medium	Not Recommended
LinkedIn	Restricted/no general API	N/A	N/A	N/A	Not Recommended (ToS/legal risk)
GitHub	Yes, free, generous	Free	Optional (higher limits if authenticated)	High	Yes — Best Overall
Reddit	Yes, free tier + manual approval gate	Free (non-commercial)	Yes, approval required	Medium (tightening trend)	Conditional Yes
Hacker News	Yes, free, keyless (2 APIs)	Free	No	High	Yes — Best Free/Best API
Substack	RSS only, official	Free	No	High	Yes
YouTube	Yes, free quota	Free (quota-limited)	API key (free)	High	Yes
Discord/Slack	Bot APIs, per-server	Free	Per-server invite	Medium	Good to Have / v2

Category 5 Recommendations

- **Best Source:** Hacker News (Algolia + Firebase APIs).
- **Best Free Source:** Hacker News.
- **Most Reliable:** GitHub.
- **Easiest Integration:** Hacker News Algolia API (single GET, JSON, no auth).
- **Best API:** GitHub API (breadth) tied with HN API (simplicity).
- **Best for MVP:** GitHub + Hacker News + YouTube Data API (quota-limited) + Substack RSS. **Explicitly exclude X and LinkedIn.** Treat Reddit as optional/best-effort.
- **Overall Recommendation:** This category should be renamed internally as “Developer & Community Signal” rather than “Social Intelligence” — the two most

valuable and safest sources (GitHub, HN) are developer-community platforms, not mainstream social networks, and the mainstream networks (X, LinkedIn) are either paywalled or legally risky to scrape.

CATEGORY 6 — RESEARCH PAPERS

This category has the strongest free-API coverage of any category in the report.

Source	API	Cost	Auth	Coverage	Notes
arXiv	Yes — official, free, keyless (<code>export.arxiv.org/api/query</code> , plus OAI-PMH bulk-metadata harvesting)	Free	No key required (rate-limit via delay between requests, per arXiv's API ToS)	Preprints across CS/AI/ML/physics/math	Best Source for pre-publication AI/ML research; extremely high recall for anything ML-related since nearly every ML paper is posted here first
Semantic Scholar	Yes — official, free API (<code>api.semanticscholar.org</code>), API key free to obtain and raises rate limits	Free (with optional free key)	Optional key	Cross-disciplinary, citation graphs, TLDR summaries	Excellent for citation-count/influence ranking, not just discovery
OpenAlex	Yes — official, fully free, no key required at all , extremely generous (100,000 calls/day documented historically)	Free	No	Broadest open scholarly graph (successor to Microsoft Academic Graph)	Recommended as the primary structured metadata backbone — covers papers, authors, institutions, funders, concepts, in one graph
CrossRef	Yes — official, free, keyless (polite-pool with an email header for higher limits)	Free	No (email header recommended)	DOI-centric metadata for nearly all formally published papers	Best for resolving DOIs and publisher metadata
CORE	Yes — official, free API (registration for a key required)	Free	Key required (free)	Open-access aggregator, full-text search	Good supplementary full-text source

Source	API	Cost	Auth	Coverage	Notes
				across repositories	
Google Scholar	No official API. Google has never published one, and third-party scraping tools (e.g., SerpApi wrappers) explicitly exist because there is no legitimate free path.	N/A	N/A	Broadest coverage, but access-restricted	Not Recommended as an automated source — no compliant free API exists; use OpenAlex/Semantic Scholar instead, which cover most of the same ground legitimately
Papers with Code	Discontinued. Papers With Code was retired by Meta in July 2025; the domain now redirects to Hugging Face’s Trending Papers feature, and the SOTA leaderboards are no longer maintained.	N/A	N/A	N/A	Do not build against this — it no longer exists as an active product. Use Hugging Face Papers / Trending Papers (huggingface.co/papers) as the spiritual successor; a frozen historical dataset remains on GitHub (paperswithcode/paperswithcode-data) for backfill only.
PubMed / PubMed Central (E-utilities)	Yes — official, free, keyless (key optional, raises rate limit)	Free	Optional key	Biomedical; relevant only for AI-in-healthcare research tracking	Good to Have, niche relevance
University institutional repositories	No unified API; each institution runs its own DSpace/EPrints instance, most of which expose OAI-PMH	Free	No	Long-tail coverage	Low priority; OpenAlex/CORE already aggregate most of this

Category 6 Recommendations

- **Best Source:** arXiv API (for AI/ML specifically — this is where the field publishes first).
 - **Best Free Source:** Tie between arXiv and OpenAlex.
 - **Most Reliable:** OpenAlex (backed by OurResearch, a well-funded nonprofit successor to Microsoft Academic Graph, with strong API-stability track record).
 - **Easiest Integration:** OpenAlex (no key, clean REST/JSON, excellent docs).
 - **Best API:** OpenAlex, for breadth; arXiv, for AI/ML-specific recall.
 - **Best for MVP:** arXiv API (new-paper firehose, filtered to cs.AI/cs.CL/cs.LG/cs.CV categories) + OpenAlex (citation/impact enrichment) + Hugging Face Papers (community-relevance signal, replacing the dead Papers with Code).
 - **Overall Recommendation:** This is the single strongest, cheapest, most future-proof category in the report — build it early, since all core sources are official, free, and keyless or near-keyless.
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CATEGORY 7 — UNIVERSITIES & RESEARCH LABS

There is no unified API for “what universities are publishing/announcing.” The practical, free architecture is:

Approach	Mechanism	Cost	Notes
Lab-level RSS/blogs	Stanford HAI, Stanford AI Lab (SAIL), MIT CSAIL, Berkeley BAIR, CMU ML Blog, Oxford/Cambridge AI groups — most maintain WordPress/Ghost blogs with RSS	Free	Verify per-lab; university blogs are generally RSS-friendly since they run on standard CMS platforms
Corporate AI labs	Google DeepMind, Meta FAIR, Microsoft Research (MSR has a confirmed strong publication/blog RSS tradition), NVIDIA Research	Free (RSS/scraping)	Overlaps with Category 2; do not duplicate ingestion pipelines
Government research orgs	DARPA, NSF (AI-related programs), India’s DST/CSIR, EU’s Horizon Europe/ERC	Free, official (see Category 8 for government-specific detail)	Lower frequency, high policy relevance
Paper-level attribution (recommended primary mechanism)	Route this entire category through the Category 6 pipeline — OpenAlex and arXiv both carry institution-affiliation metadata per	Free	This is the key architectural recommendation for this category

Approach	Mechanism	Cost	Notes
	paper, so “what is Stanford publishing” becomes a filter on the research-papers pipeline rather than a separate scraper per university		

Category 7 Recommendations

- **Best Source:** OpenAlex filtered by institution ID (via the Category 6 pipeline) — avoids building 50+ bespoke university scrapers.
- **Best Free Source:** Same.
- **Easiest Integration:** OpenAlex institution filter (<https://api.openalex.org/works?filter=institutions.id:...>).
- **Best for MVP:** Do not build dedicated university scrapers at all in v1. Derive “university/lab activity” as a downstream view over the arXiv + OpenAlex data already being ingested for Category 6, plus a short curated RSS list (~10-15 feeds) for the handful of labs whose announcements matter most (Stanford HAI, MIT CSAIL, Berkeley BAIR, MSR, DeepMind, FAIR).
- **Overall Recommendation:** Treat this as a **derived/virtual category**, not a new ingestion pipeline — the highest-leverage engineering decision in this whole report.

CATEGORY 8 — GOVERNMENT POLICIES

Official sources only, as required. All are free by nature (public-sector transparency mandates).

Jurisdiction	Source	Mechanism	Notes
India	NITI Aayog (niti.gov.in) — AI policy documents, National AI Strategy	No API; document/press-release page, some RSS on government press portals	Primary Indian AI-policy source
India	IndiaAI Mission (indiaai.gov.in)	Website with news/reports section; no confirmed public API	Tracks India’s national AI compute/funding mission
India	PIB (Press Information Bureau) (pib.gov.in)	PIB does provide RSS feeds per ministry, a well-known and durable official	Recommended primary

Jurisdiction	Source	Mechanism	Notes
		mechanism for Indian government announcements	mechanism for India-wide policy tracking
India	Startup India portal (startupindia.gov.in)	No general public API confirmed for the startup registry; recognition-certificate lookups exist but are not built as an open dataset API	Use for manual/periodic reference; not a real-time feed source
USA	NIST AI (AI Risk Management Framework, US AI Safety Institute)	Official site, publishes documents; NIST also has broader open-data APIs for other domains but AI-policy content itself is document-based, not API-based	High-trust primary source
USA	whitehouse.gov / Federal Register	Federal Register has an official, free, well-documented REST API (federalregister.gov/developers/api/v1) — searchable by agency, topic (e.g., “artificial intelligence”), date	Best Government API in this entire report — official, free, structured, stable, covers executive orders and agency rules
Singapore	IMDA / Smart Nation / Singapore’s National AI Strategy (smartnation.gov.sg, imda.gov.sg)	Website/document based; no confirmed general API	High-trust primary source
EU	European Commission Digital Strategy / EU AI Act portal (digital-strategy.ec.europa.eu)	Document-based; EU also runs an open-data portal (data.europa.eu) with API access for structured datasets, though the AI Act text itself is a legal document, not an API resource	Use EUR-Lex (eur-lex.europa.eu) for the official legal text, which has a

Jurisdiction	Source	Mechanism	Notes
			documented web service for legal-document retrieval
UK	UK Government AI policy hub (gov.uk/government/organisations/department-for-science-innovation-and-technology)	gov.uk publishes official Atom/RSS feeds for most content types , a long-standing and reliable UK government mechanism	Recommended — RSS is genuinely first-class on gov.uk
UAE	UAE AI Office / Ministry of AI	Website/document based	Verify for RSS at implementation time
Saudi Arabia	SDAIA (Saudi Data & AI Authority)	Website/document based	Verify for RSS at implementation time
OECD	OECD.AI Policy Observatory (oecd.ai)	OECD.AI maintains a structured policy database with a documented interface for querying national AI policies/strategies across member and partner countries — the closest thing to a genuine cross-country “AI policy API” that exists today	Recommended as the best cross-jurisdictional source in this category — verify current API terms at implementation time, as OECD.AI’s technical access layer has evolved over the past few years

Category 8 Recommendations

- **Best Source:** OECD.AI Policy Observatory (cross-country structured data).
 - **Best Free Source:** All sources in this category are free by definition (government transparency).
 - **Most Reliable:** Federal Register API (US) — genuinely production-grade government API engineering.
 - **Easiest Integration:** Federal Register API and gov.uk RSS.
 - **Best API:** Federal Register API.
 - **Best RSS:** gov.uk (UK) and PIB (India).
 - **Best for MVP:** Federal Register API (US) + gov.uk RSS (UK) + PIB RSS (India) + OECD.AI (cross-country) + manual quarterly review of NITI Aayog/IndiaAI/EU AI Act portal (lower-frequency, document-heavy sources not worth real-time polling).
 - **Overall Recommendation:** This category is inherently low-frequency and document-centric. Do not over-engineer real-time polling; a weekly batch job per source is sufficient and matches how governments actually publish.
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CATEGORY 9 — OPEN SOURCE ECOSYSTEM

The strongest, most stable, most free-API-rich category alongside Research Papers.

Source	API	Cost	Auth	Notes
GitHub REST/GraphQL API	Yes, official, free	Free (paid tiers exist for orgs, irrelevant here)	Personal Access Token recommended (free); 5,000 req/hr authenticated vs. 60/hr unauthenticated — standard, stable limits	The single most important API in this entire report. Covers repos, releases, trending signal (via search API sorted by stars/recently-created), commit activity, issues
GitHub Trending	No official API for the Trending page specifically, but well-established open-source	Free	Search API needs auth for higher	Recommended building trending

Source	API	Cost	Auth	Notes
	community wrappers exist that scrape it responsibly; alternatively, replicate “trending” logic using the official Search API (<code>?q=created:>DATE&sort=stars</code>)		limits	logic on the official Search API rather than depending on scraping the Trending HTML page, for durability
Hugging Face Hub API	Yes, official, free, keyless for public read	Free	No	Already profiled in Category 2; also core to this category (models + datasets + Spaces)
PyPI	Yes — official JSON API per-package (<code>pypi.org/pypi/{package}/json</code>) + PyPI’s BigQuery public dataset for aggregate download stats	Free	No	Good for tracking new/trending Python AI packages
npm	Yes — official registry API (<code>registry.npmjs.org</code>), plus npm download-count API (<code>api.npmjs.org/downloads</code>)	Free	No	JS/TS AI-tooling ecosystem tracking
Product Hunt	Yes — official GraphQL API v2 (<code>api.producthunt.com/v2/api/graphql</code>)	Free for non-commercial use — the Product Hunt API must not be used for commercial purposes	OAuth access token required (free to obtain)	Rate limits: 6,250 complexity points per 15 minutes on the GraphQL endpoint, and 450 requests per 15 minutes on REST-style

Source	API	Cost	Auth	Notes
		without contacting them directly		v2 endpoints — generous for a dashboard use case
GitLab	Yes, official, free, well-documented REST/GraphQL API (GitLab is API-first by design)	Free	Token for higher limits	Secondary to GitHub but useful for projects that primarily live on GitLab
Apache Software Foundation	No unified API; individual project mailing-list/announce feeds vary; ASF publishes some project data via its own infrastructure	Free (RSS per project where available)	No	Low priority; relevant AI projects (e.g., some data-infra tools) better tracked via their GitHub mirrors
Linux Foundation / LF AI & Data	Website/news-based; some project-level GitHub tracking overlaps with the GitHub API pipeline	Free	No	Low-frequency source
CNCF	CNCF publishes a Landscape dataset (open, on GitHub: github.com/cncf/landscape) that is structured and free	Free	No	Relevant mainly for AI-infra/MLOps tooling that's part of the cloud-native landscape
Awesome Lists (awesome-*)	No API beyond GitHub's own (they're just repos)	Free	Via GitHub API	High-curation-value, low-

Source	API	Cost	Auth	Notes
GitHub repos)				engineering -cost: treat as a periodically -refreshed curated-link source rather than a live feed

Category 9 Recommendations

- **Best Source:** GitHub REST/GraphQL API.
- **Best Free Source:** GitHub API (breadth) and Hugging Face Hub API (AI-specific depth) tied.
- **Most Reliable:** GitHub API — arguably the most stable, well-governed public API of any source in this entire report, with a decade-plus track record.
- **Easiest Integration:** PyPI/npm JSON endpoints (no auth needed at all for public read).
- **Best API:** GitHub API.
- **Best for MVP:** GitHub Search/Repos/Releases API (trending + release tracking across curated AI org list) + Hugging Face Hub API (models/datasets/spaces) + PyPI/npm JSON endpoints for a curated list of AI-tooling package names.
- **Best for Future Scaling:** Add Product Hunt API once the dashboard needs product-launch discovery, since it requires OAuth setup that's slightly more friction than the keyless APIs above.
- **Overall Recommendation:** This category, combined with Research Papers (Category 6), should be built first — both are dominated by official, free, stable, well-documented APIs with no commercial gatekeeping, making them the highest-ROI, lowest-risk starting point for the entire dashboard.

CATEGORY 10 — DEVELOPER ECOSYSTEM

Source	API	Cost	Auth	Notes
Stack Overflow	Yes — official Stack Exchange API (api.stackexchange.com), free, generous quota (historically 10,000 requests/day per IP without a key, 300/day without registering an app — verify current quotas at build time as	Free	Optional key raises quota	Good for gauging developer pain- points/question- volume trends around specific AI tools/frameworks

Source	API	Cost	Auth	Notes
	Stack Exchange adjusts these periodically)			
Dev.to	Yes — official, free, keyless Forem API (dev.to/api)	Free	No (optional key for write actions, irrelevant here)	Clean JSON API, good AI-tag coverage
Hashnode	Yes — official GraphQL API (gql.hashnode.com), free	Free	No key required for public reads	Similar role to Dev.to
Medium (publications)	No general public API for content since Medium deprecated much of its open API surface years ago; RSS still works per-publication/per-user (medium.com/feed/@user or /feed/publication-name)	Free	No	RSS is the durable path here, not API
Engineering blogs (Netflix, Uber, Airbnb, Stripe, etc.)	No unified API; nearly all run on platforms with native RSS (WordPress, Ghost, custom-with-feed)	Free	No	Curate a list of ~15-20 relevant engineering blogs relevant to AI infra/tooling
Cloud providers' "What's New" feeds	AWS (confirmed official RSS, see Category 2), Google Cloud (release notes have RSS for many products), Azure (has RSS for updates)	Free	No	Track AI/ML-relevant product-service SKUs specifically, not the entire firehose
Framework releases (PyTorch, TensorFlow, LangChain, LlamaIndex, JAX, Transformers)	All are GitHub-hosted → covered natively by the GitHub Releases API already built in Category 9	Free	Via GitHub API	No separate integration needed — this is a filter on the existing GitHub pipeline, not a new source
Developer newsletters (e.g., Pointer, Console.dev, Software Lead)	RSS where self-hosted or Substack-hosted	Free	No	Supplementary; lower priority than the structured APIs above

Source	API	Cost	Auth	Notes
Weekly)				

Category 10 Recommendations

- **Best Source:** Stack Exchange API (structured signal on developer pain points/trends).
- **Best Free Source:** Dev.to API (cleanest free, keyless JSON API in this category).
- **Most Reliable:** GitHub Releases API (via the Category 9 pipeline) for framework-release tracking specifically.
- **Easiest Integration:** Dev.to API.
- **Best API:** Stack Exchange API.
- **Best for MVP:** Filter the existing GitHub pipeline (Category 9) for framework repos (PyTorch, Transformers, LangChain, etc.) + Dev.to API (AI-tagged posts) + AWS/GCP/Azure “what’s new” RSS filtered to AI/ML services. Treat engineering blogs and newsletters as a v2 curated-RSS layer.
- **Overall Recommendation:** Most of this category’s value is a **derived view over Category 9’s GitHub pipeline**, not a separate ingestion system — a second instance of the “virtual category” pattern found in Category 7.

MASTER COMPARISON TABLE

Ranked by (1) cost-to-value ratio, (2) reliability, (3) technical ease, consistent with the project’s stated optimization priorities (max coverage, max reliability, min cost).

Rank	Source	Category	Type	Cost	Reliability	Difficulty	MVP Tier
1	GitHub REST/GraphQL API	Open Source / Model Releases / Dev Ecosystem	Official API	Free	High	Low	Must Have
2	Hugging Face Hub API	Model Releases / Open Source	Official API	Free	High	Low	Must Have
3	arXiv API	Research Papers	Official API	Free	High	Low	Must Have
4	OpenAlex API	Research Papers / Universit	Official API	Free	High	Low	Must Have

Rank	Source	Category	Type	Cost	Reliability	Difficulty	MVP Tier
		ies					
5	Hacker News (Firebase + Algolia APIs)	Social Intelligence / AI News	Official API	Free	High	Low	Must Have
6	TechCrunch RSS (+ Verge/Ars Technica)	AI News	Official RSS	Free	High	Low	Must Have
7	SEC EDGAR Form D full-text search	Startup Funding	Official API/Search	Free	High	Medium	Must Have
8	Federal Register API	Government Policy (US)	Official API	Free	High	Low	Must Have
9	Substack RSS (curated newsletters)	AI News / VC Intelligence	Official RSS	Free	Medium-High	Low	Must Have
10	Semantic Scholar API	Research Papers	Official API	Free	High	Low	Good to Have
11	CrossRef API	Research Papers	Official API	Free	High	Low	Good to Have
12	PyPI / npm JSON endpoints	Open Source	Official API	Free	High	Low	Good to Have
13	Dev.to API	Developer Ecosystem	Official API	Free	High	Low	Good to Have
14	Stack Exchange API	Developer Ecosystem	Official API	Free (quota)	High	Low	Good to Have
15	YouTube Data API v3	Social Intelligence	Official API	Free (quota)	High	Medium	Good to Have

Rank	Source	Category	Type	Cost	Reliability	Difficulty	MVP Tier
16	gov.uk RSS	Government Policy (UK)	Official RSS	Free	High	Low	Good to Have
17	PIB RSS	Government Policy (India)	Official RSS	Free	High	Low	Good to Have
18	OECD.AI Policy Observatory	Government Policy (cross-country)	Structured DB	Free	Medium-High	Medium	Good to Have
19	AWS "What's New" RSS	Model Releases / Dev Ecosystem	Official RSS	Free	High	Low	Good to Have
20	Product Hunt API v2	Open Source / Startup Funding	Official API	Free (non-commercial)	High	Medium	Good to Have
21	Reddit API (free tier)	Social Intelligence	Official API	Free, approved	Medium (tightening)	Medium-High	Experimental
22	Company blogs w/o RSS (OpenAI, Anthropic, DeepMind, Meta AI)	AI News / Model Releases	Scraping	Free	High (content) / Medium (pipeline)	Medium	Must Have (accept scraping risk)
23	University lab RSS (curated list)	Universities & Research	Official RSS	Free	Medium-High	Low	Good to Have
24	CORE API	Research Papers	Official API	Free (key)	Medium-High	Low	Future Expansion
25	Startup India /	Government /	Manual/scrape	Free	Medium	Medium	Future Expansion

Rank	Source	Category	Type	Cost	Reliability	Difficulty	MVP Tier
	IndiaAI portals	Startup Funding					
26	Crunchbase	Startup Funding	Official API	Paid — \$49-199/mo self-serve; enterprise-only for full API, \$50K+/yr	High	Medium	Not Recommended for MVP
27	Dealroom / Tracxn / PitchBook	Startup Funding	Official API	Paid, enterprise	High	Medium	Not Recommended
28	X (Twitter) API	Social Intelligence	Official API	Paid only	Medium	Medium	Not Recommended
29	LinkedIn (scraping)	Social Intelligence	Unofficial/ ToS-violating	N/A	N/A	N/A	Not Recommended (legal risk)
30	NewsAPI.org / general news aggregators	AI News	Official API	Free tier dev-only; \$449+/mo for production	High	Low	Not Recommended (RSS is sufficient and free)

FREE VS. PAID COMPARISON

Dimension	Free-Tier-Sufficient Path	Where Paid Becomes Necessary
AI News	Fully covered by RSS + HN + Substack	Never — no paid upgrade needed
Model Releases	Fully covered by HF Hub API + GitHub API	Never — no paid upgrade needed
Startup Funding	Partially covered (SEC EDGAR + RSS + gov portals); misses private/seed-stage rounds not	Crunchbase Pro/Business (\$49-199/mo) closes some gaps; full structured API access remains

Dimension	Free-Tier-Sufficient Path	Where Paid Becomes Necessary
	publicly reported and global coverage outside major markets	enterprise-priced (\$50K+/yr) regardless of budget tier
VC Intelligence	Fully covered by RSS/newsletters for narrative signal	Structured “which fund invested in what, when” data has no good free API; remains a scraping/manual-curation problem even at scale
Social Intelligence	GitHub + HN + YouTube + Dev.to cover developer-community signal well	X and LinkedIn require paid/partner access indefinitely — treat as permanently out of scope unless budget materially increases
Research Papers	Fully covered, free, indefinitely	Never — this category has no paid-tier need at any scale
Universities & Labs	Fully covered as a derived view over Research Papers pipeline	Never
Government Policy	Fully covered, free by definition	Never
Open Source Ecosystem	Fully covered, free, indefinitely	Never
Developer Ecosystem	Fully covered, free, indefinitely	Never

Key finding: 8 of 10 categories can be built entirely on free, official, stable sources with no realistic future need to go paid. Only **Startup Funding Intelligence** and parts of **Social Intelligence (X/LinkedIn)** have a genuine free-tier ceiling — and in both cases, the paid alternatives are expensive enough (\$50K+/yr for full Crunchbase API access; X’s enterprise API pricing is comparably steep) that they should be treated as “may never be worth it” rather than “upgrade when budget allows.”

MVP RECOMMENDATION (Budget: ~₹0)

The following source set delivers maximum coverage at zero cost, using only official APIs and RSS feeds:

Tier 1 (build first — all official, free, keyless or near-keyless APIs): 1. GitHub REST/Search/GraphQL API 2. Hugging Face Hub API 3. arXiv API 4. OpenAlex API 5. Hacker News (Firebase + Algolia)

Tier 2 (build second — RSS aggregation layer): 6. TechCrunch, The Verge, Ars Technica category RSS feeds 7. Curated Substack newsletters (5-8 feeds: Import AI, The Batch, Ben’s Bites, TLDR AI, plus 2-3 VC/startup newsletters) 8. AWS “What’s New” RSS 9. gov.uk and PIB (India) RSS

Tier 3 (build third — light-auth official APIs): 10. Semantic Scholar API (free key) 11. Dev.to API 12. Stack Exchange API 13. YouTube Data API v3 (free Google Cloud key) 14. Federal Register API 15. PyPI/npm JSON endpoints

Tier 4 (build fourth — higher-maintenance but still free): 16. Scraping layer for OpenAI/Anthropic/DeepMind/Meta AI blogs (no RSS exists) 17. SEC EDGAR Form D search for US funding signal

Explicitly excluded from MVP: Crunchbase, Dealroom, Tracxn, PitchBook, X/Twitter API, LinkedIn (any form), general-purpose paid news aggregators.

All of Tier 1–3 route naturally into a Gemini-powered summarization/classification/deduplication layer (the subject of Deliverable 3), and this source mix already delivers meaningful coverage of all 10 requested categories without a single paid contract.

SCALING RECOMMENDATION (When Budget Increases)

Trigger	Add
Need deeper, structured startup-funding coverage beyond what SEC EDGAR + RSS provide	Crunchbase Pro/Business (\$49-199/mo) for UI-level lookups; full API access still requires a separate enterprise conversation — budget for that only if the dashboard's core value proposition becomes funding-data completeness
Need India/APAC-specific funding depth	Evaluate Tracxn (has stronger India/APAC coverage historically than Crunchbase) once budget allows
Need X/Twitter sentiment or breaking-news signal	Only once budget can absorb X's enterprise API pricing; until then, treat X coverage as “whatever gets reported secondhand by TechCrunch/HN”
Reddit approval denied or revoked at free tier	Consider a modest paid Reddit commercial tier only if Reddit-specific community sentiment becomes a core product feature, not a nice-to-have
News-API scraping maintenance burden becomes too high	Revisit a mid-tier paid news aggregator (e.g., NewsData.io or GNews at \$24-84/mo) as a maintenance-reduction purchase, not a coverage purchase — RSS already provides the coverage
Need premium engineering-blog/dev-newsletter breadth	Low priority even at scale — RSS continues to serve this well; unlikely to ever require a paid tier

INTEGRATION PRIORITY CLASSIFICATION

Must Have: GitHub API, Hugging Face Hub API, arXiv API, OpenAlex API, Hacker News API, TechCrunch/Verge/Ars RSS, Substack RSS (curated), SEC EDGAR, Federal Register API, company-blog scraping (OpenAI/Anthropic/DeepMind/Meta AI).

Good to Have: Semantic Scholar, CrossRef, Dev.to, Stack Exchange, YouTube Data API, gov.uk RSS, PIB RSS, OECD.AI, AWS RSS, Product Hunt API, PyPI/npm endpoints, curated university-lab RSS.

Future Expansion: Crunchbase (if budget allows), Tracxn/Dealroom (India/APAC depth), CORE API, expanded government-portal coverage (Startup India, IndiaAI, EU AI Act portal), Discord/Slack community bots.

Experimental: Reddit API (approval-gated, trending toward more restrictive), any LinkedIn integration (legal risk — recommend never moving this out of “experimental,” i.e., effectively excluded), X/Twitter (cost-gated indefinitely).

FINAL RECOMMENDATION — LEAD ARCHITECT’S DECISION

If I were building the MVP data layer for this dashboard today, I would launch with exactly the Tier 1–3 source set defined above, for four reasons:

1. **Every Tier 1–3 source is an official, sanctioned, free integration point** — no ToS risk, no scraping fragility (except the deliberate, isolated exception of four company blogs where no RSS exists and the content value justifies the maintenance cost).
2. **Two categories — Research Papers and Open Source Ecosystem — are essentially “solved” by free APIs alone**, and both are unusually high-signal for an AI/tech intelligence dashboard specifically. I would build these first, since they will make the dashboard immediately useful even before funding/VC/social layers are added.
3. **The one category with a real free-tier ceiling (Startup Funding) still has a legitimate, free, primary-source option in SEC EDGAR** that most consumer dashboards ignore in favor of expensive aggregators — Form D filings are the actual legal record of most VC-backed raises in the US, and pairing that with RSS-based narrative coverage gets 80% of Crunchbase’s practical value at 0% of its cost.
4. **Two entire categories (Universities & Research Labs, and much of Developer Ecosystem) don’t need dedicated ingestion pipelines at all** — they’re better built as filtered/derived views over the Research Papers and Open Source pipelines respectively. This is the single highest-leverage architectural decision in this report, and it should carry directly into Deliverable 2’s information architecture: fewer pipelines, more views.

I would deliberately **not** build against X, LinkedIn, or Crunchbase in v1 — not because they lack data value, but because their cost/risk profile (paid-only, ToS-risk, or enterprise-

gated) is incompatible with the ₹0 constraint and, in LinkedIn’s case, incompatible with “never recommend anything illegal” as a hard constraint on this project.

This source foundation is sufficient for Deliverable 2 (Information Architecture) to begin schema design immediately: the 15 Tier 1–3 sources above should map to a small number of ingestion pipeline “types” (structured-API-poll, RSS-poll, scrape-with-fallback), not 15 bespoke integrations — which is the architectural insight this research report is meant to hand off.

End of Deliverable 1. This report is scoped to source research only, per the brief — no dashboard, frontend, or backend architecture is proposed here; those are Deliverables 2 and 3.